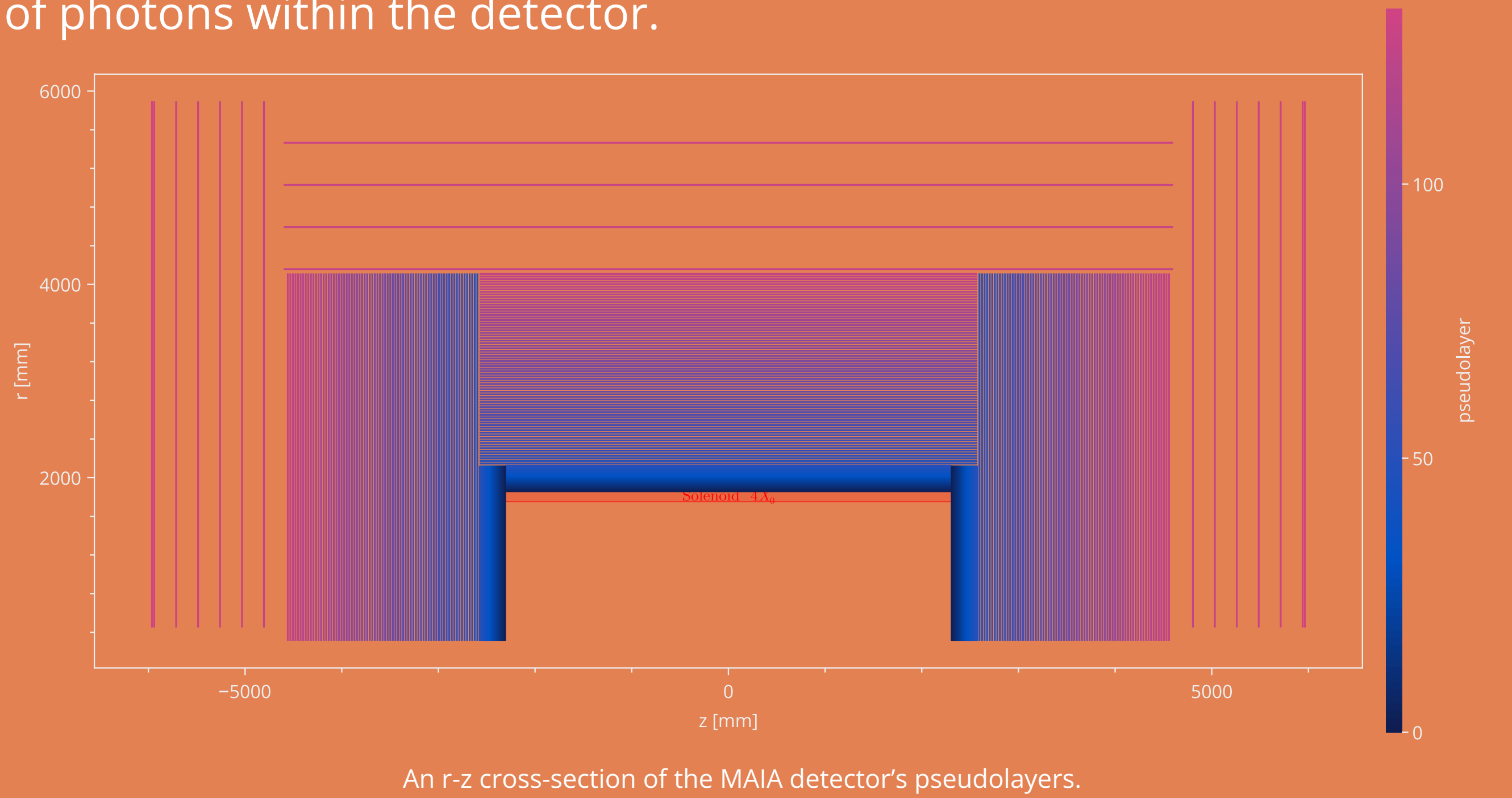


The MAIA Detector & Motivation

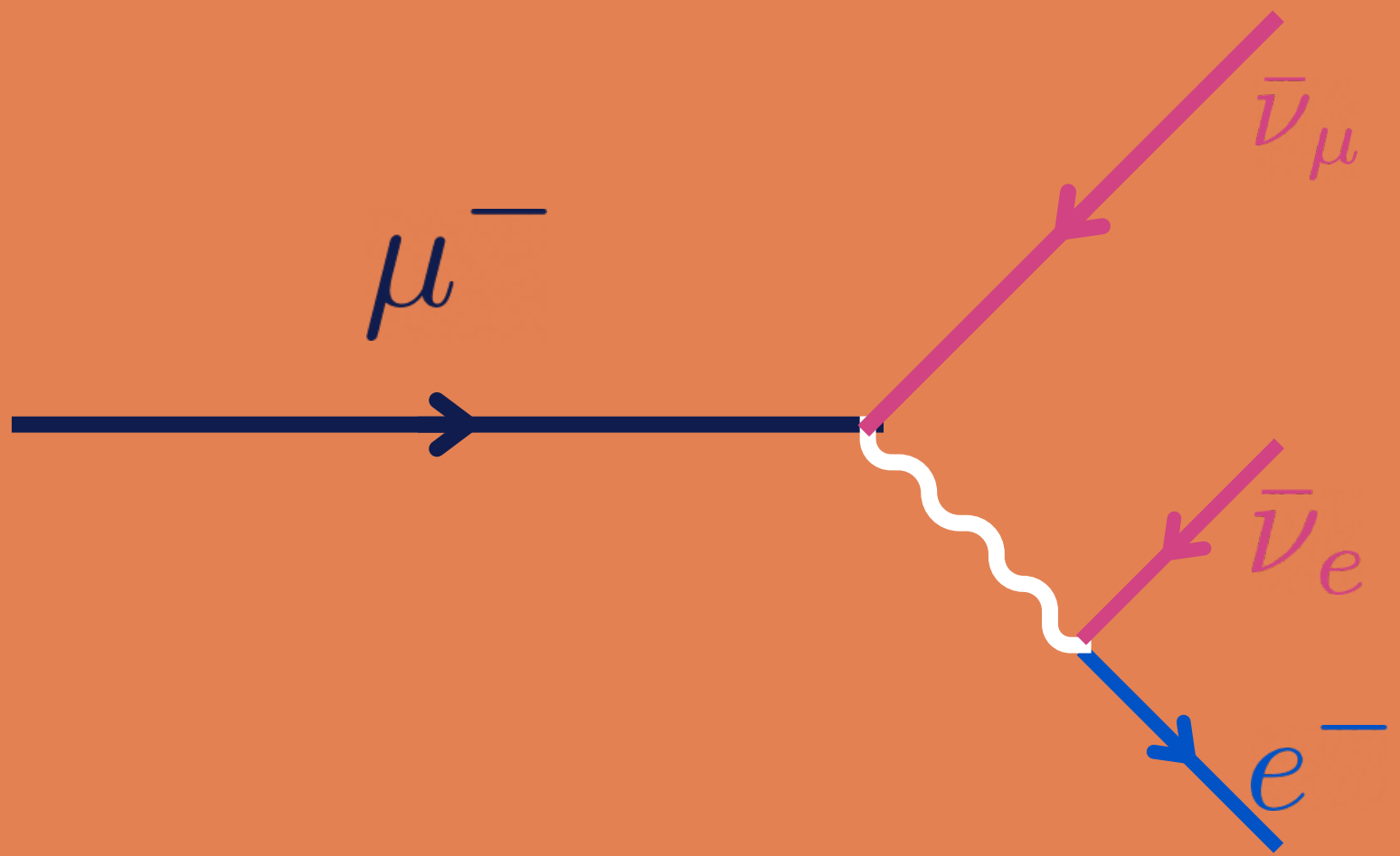
The MAIA (Muon Accelerator Instrumented Apparatus) Detector is composed of several distinct layers, each designed to reconstruct different particles. Of special interest to us is the **Electromagnetic Calorimeter**, the system designed to measure the energy of charged objects and photons. As we optimize the detector we are attempting to understand the effects our design choices have on the reconstruction of photons within the detector.



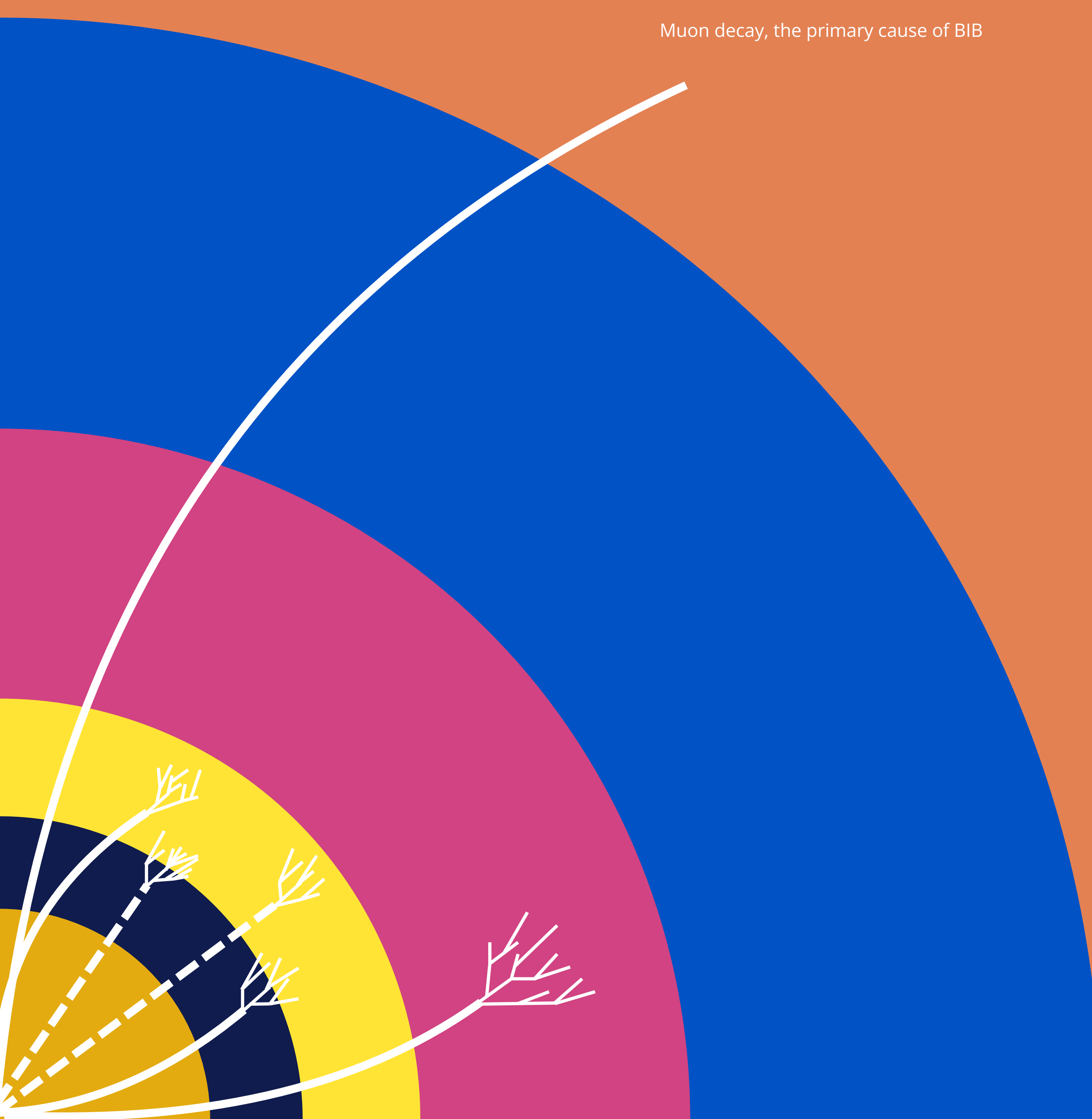
An r-z cross-section of the MAIA detector's pseudolayers.

In the MAIA detector we have placed the solenoid inside the calorimeters in order to reduce the amount of so-called **Beam-Induced-Background** the calorimeters experience.

The incoming **Muon Beam** is unstable, and as such, will decay into high energy **Electrons** and **Neutrinos** that deposit in our detector volume. These represent the largest challenge in designing an effective detector for a muon collider.



Muon decay, the primary cause of BIB



Understanding photon shower behavior at the MAIA detector

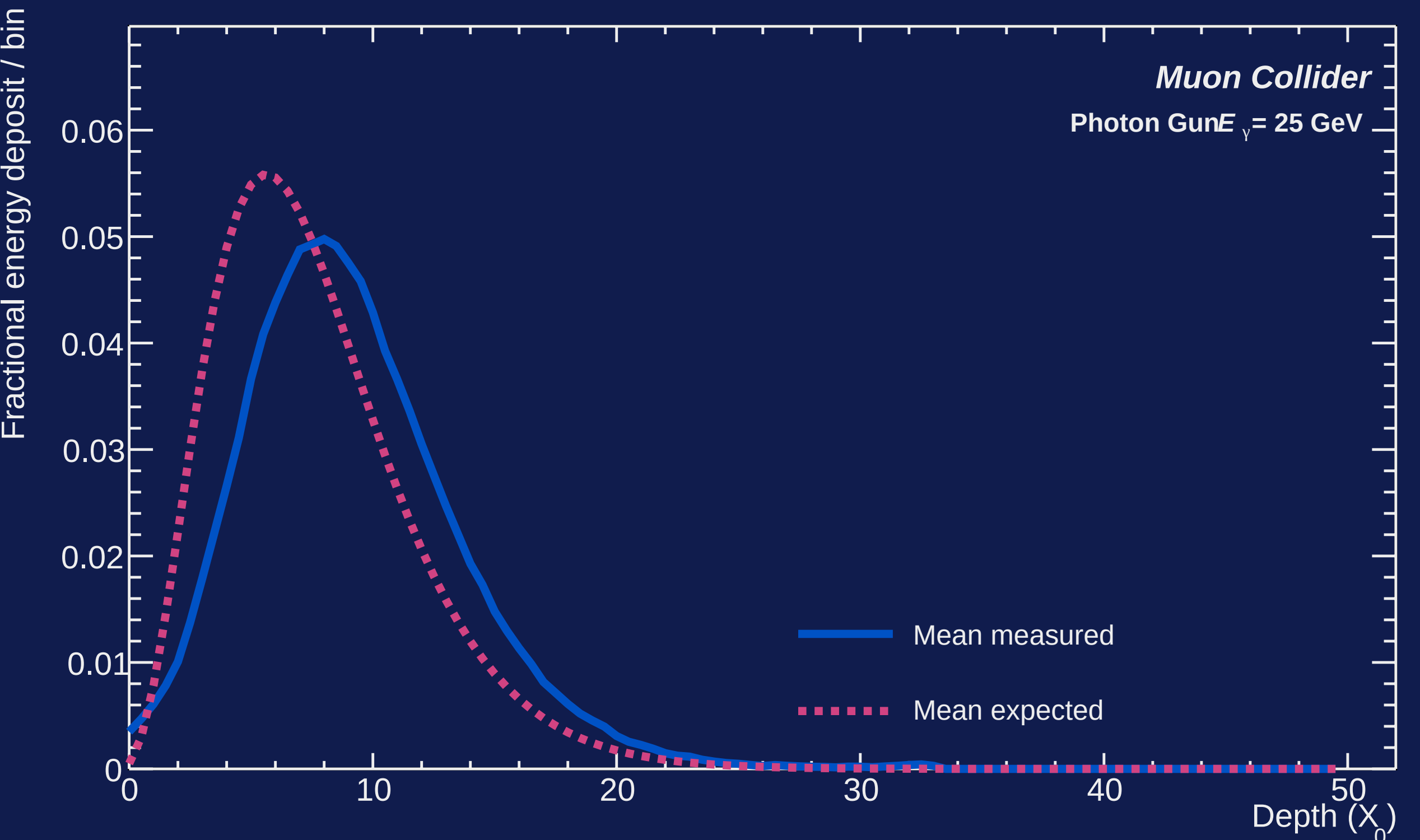
Aidan Gardner-O’Kearny¹, Tova Holmes¹, Lawrence Lee¹, The MAIA Detector Group | ¹Department of Physics, University of Tennessee

Electromagnetic Showers

The shape of an electromagnetic shower is characterized by a gamma function.

$$\frac{dE}{dt} \propto E_0 b \frac{(bt)^{a-1} e^{-bt}}{\Gamma(a)}$$

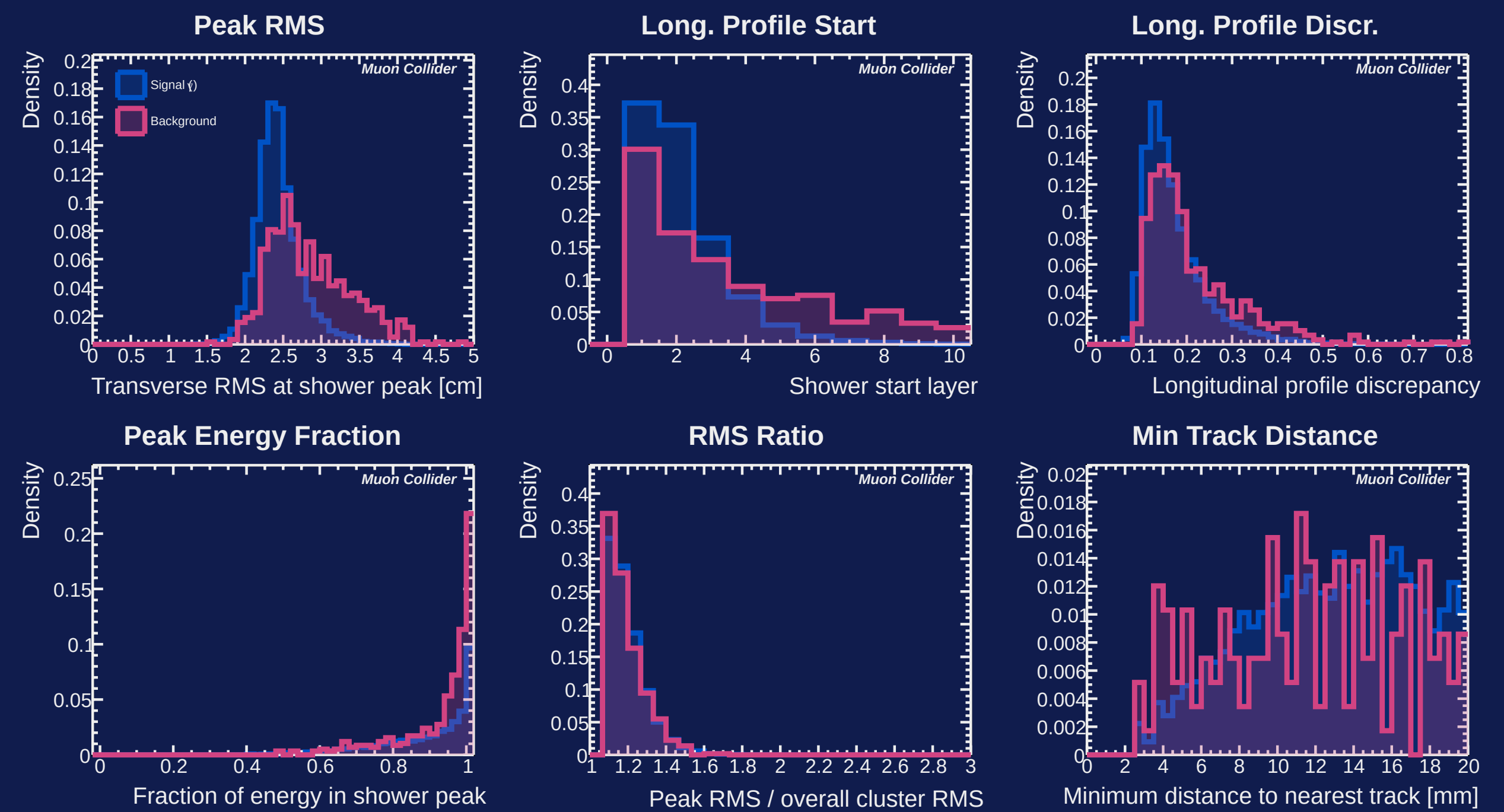
The length and peak of this curve is determined both by the material properties of the detector and by the energy of the incident particle. In this way we are able to reconstruct the energy deposition of a particle and compare it against a theoretical prediction, resulting in a metric called the **Profile Discrepancy**, a score that tells us how much like a shower an object is.



Expected and measured longitudinal shower profiles for a 25 GeV photon gun.

Photon Tagging

Inside the MAIA software, we use the **Pandora Particle Flow Algorithm** to identify physics objects within the detector. This algorithm is fed various topological variables about a cluster and provides a tag based on those variables. For photons, we use a six-dimensional classifier across twelve energy bands to distinguish a signal event from background.



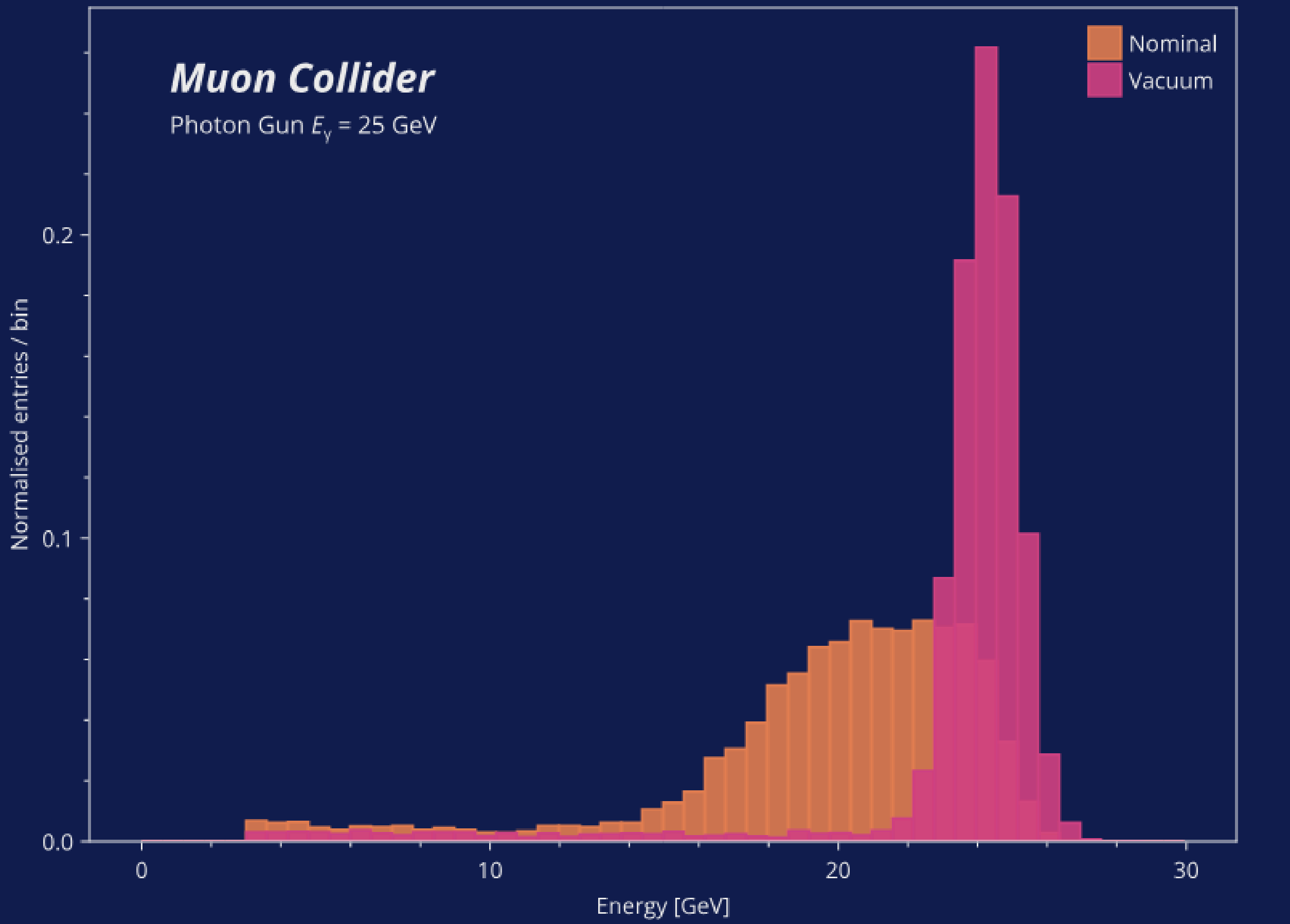
Training data used for the classifier in the highest energy bin (E>75.0 GeV)

Acknowledgements

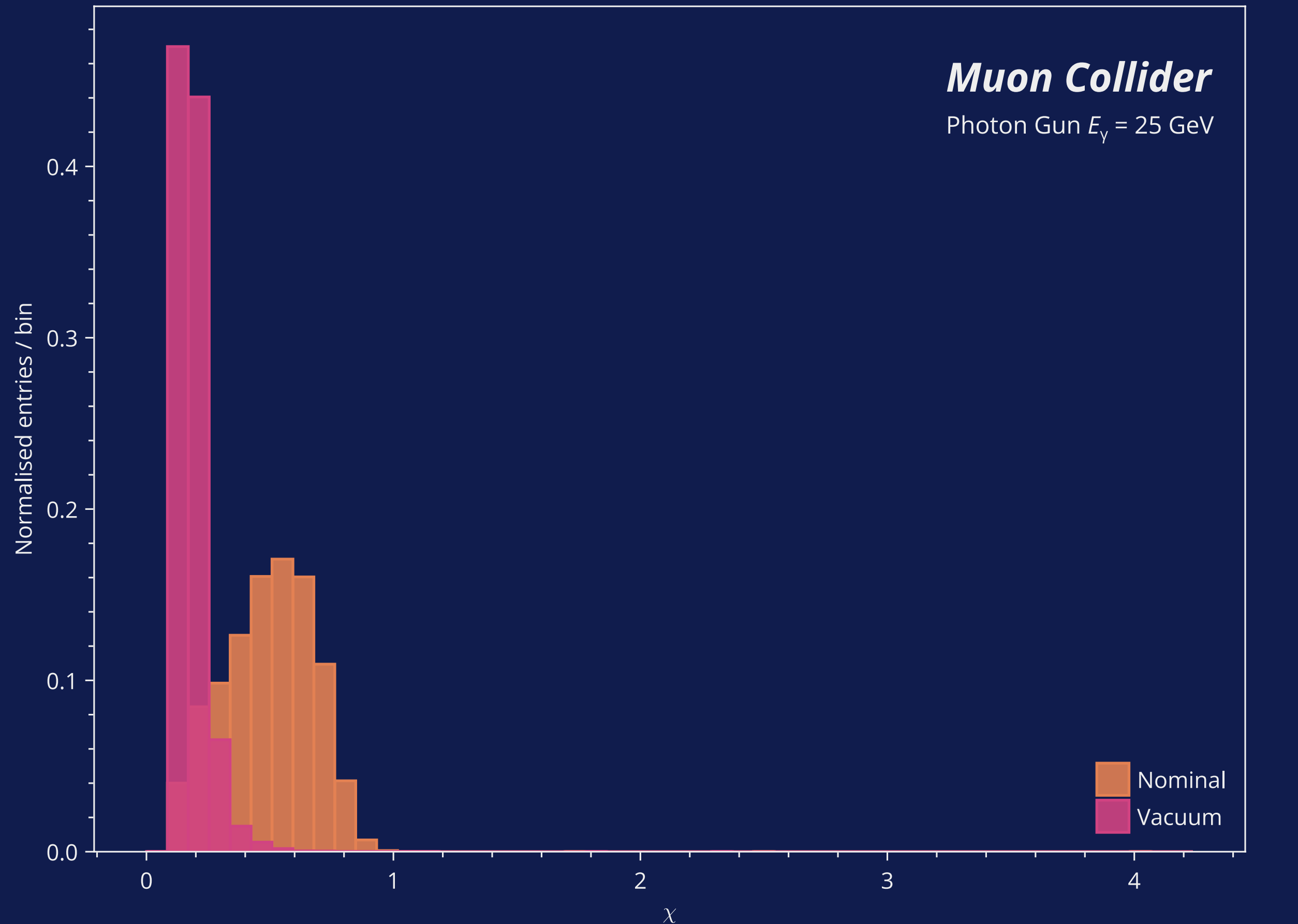
Special thanks to Tova Holmes for her guidance on this project and to Larry Lee for useful feedback. Additional thanks to JP Devran, Rithika Ganesan, Caroline Riggall, and Julian Watts for providing other sets of eyes on the poster. Funding for this project was provided by the Research Corporation for Science Advancement and the Department of Energy.

The MAIA Solenoid

While moving the solenoid inside the calorimeters is effective at shielding BIB, it also represents a large dead zone of material. We can see the effect this has on photons by simulating the passage of them through both the **Nominal** MAIA design and a version where we replace the solenoid material with **Vacuum**. As can be seen we have both worse energy resolution and discrepancy scores due to the solenoid.



The energy of photon gun events simulated in both the nominal MAIA design and a design with no solenoid material.



The profile discrepancy scores of photon gun events simulated in both the nominal MAIA design and a design with no solenoid material.

Conclusions & Next Steps

We have identified a severe issue with the reconstruction of photons within the MAIA detector. We are currently exploring ways to correct this behavior. These methods include both purely Monte Carlo driven calibration as well as more aggressive solutions such as the addition of an ATLAS-style presampler.

